

VOXELFLIGHT

Single-pass drone video to an explorable 3D scene

For missions where another flight
may not be possible.

Single-Pass Drone Video to Accurate 3D Model Generation System



ACTUAL INPUT FRAME

SIH26158

National Technical Research Organisation

Team name: [To be added] Team ID: [To be added]

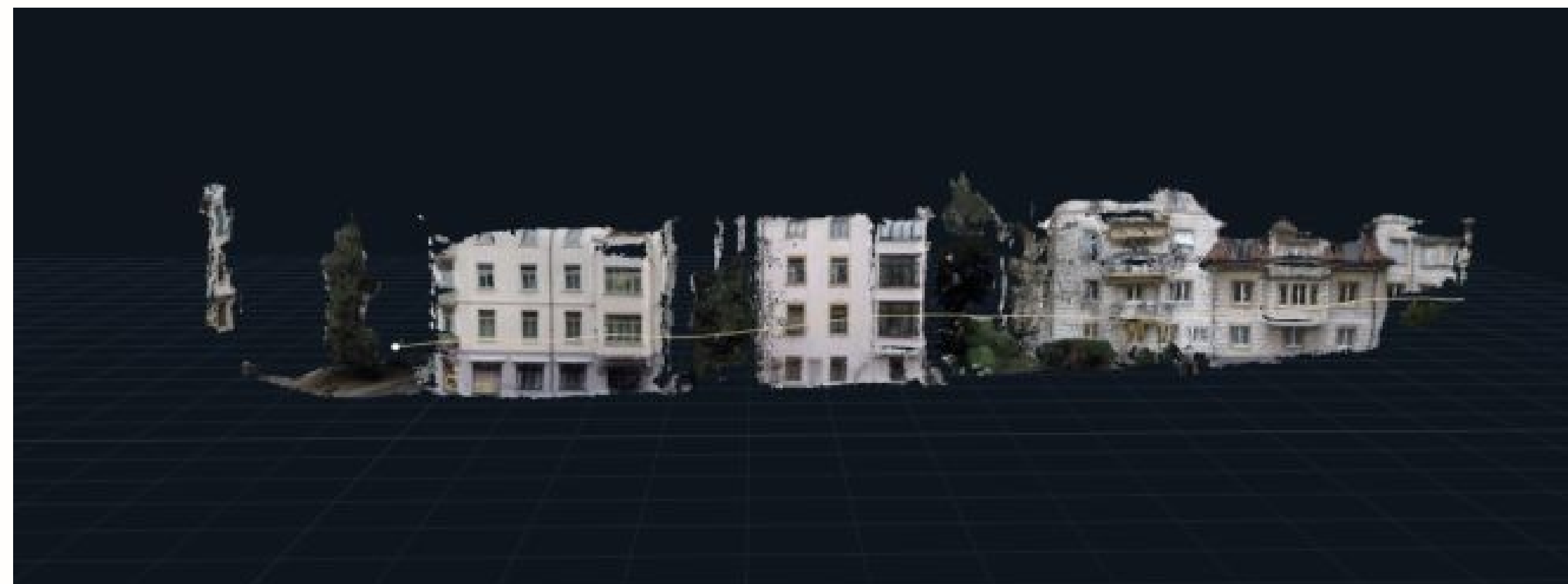
Single-pass reconstruction of visible surfaces

A video supplies overlapping views. Camera motion and predicted depth connect them into a 3D scene.

01 Flight input



02 Actual source-textured surface



Video + GPS + flight metadata

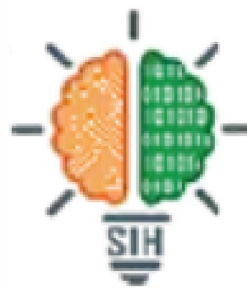
Required: 1080p or 4K video
Saved experiment: 60 s, 90 views

Optional sensors improve constraints.
No reference poses enter construction.

90 calibrated views

646,019 mesh triangles

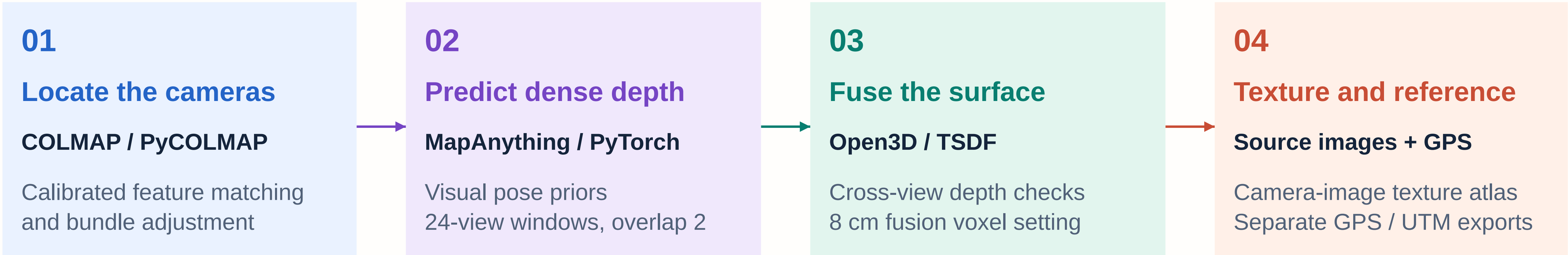
Texture comes from observed camera images. Unseen regions remain open.
Partial coverage and surface errors still require independent validation.



Calibrated reconstruction and web delivery

The saved reconstruction and default upload preview share models, but use different processing paths.

SAVED RECONSTRUCTION

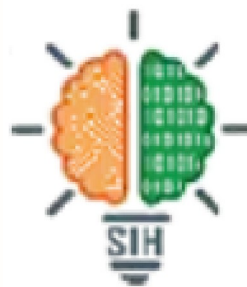


Pretrained inference. Reference trajectory enters evaluation only. Voxel size is a resolution setting.

DEPLOYED UPLOAD PREVIEW



Worker: Apple MPS. Per-user jobs require a running service. The complete saved SfM / TSDF path is separate.



Recorded results and remaining validation

Current saved reconstruction: a 60-second Zurich sequence with 90 selected views, using Apple MPS inference.

Recorded property	Result
Predicted dense points	4,208,069
Source-textured mesh	646,019 triangles
Trajectory RMSE, absolute	8.649 m
Trajectory RMSE, SE(3)	0.451 m
Trajectory RMSE, Sim(3)	0.397 m
Separate stage times, summed	6 min 10 s

Trajectory alignment changes the question

SE(3) fits rotation and translation. Sim(3) also fits scale. Neither measures surface error. The summed timing excludes upload, queue delay and packaging.

Spatial accuracy ≤ 1 m

Surface accuracy remains unmeasured. Absolute trajectory error is 8.649 m.

Entire visible scene

Coverage remains partial, with holes and fusion artifacts.

<15 min for 10 min of video

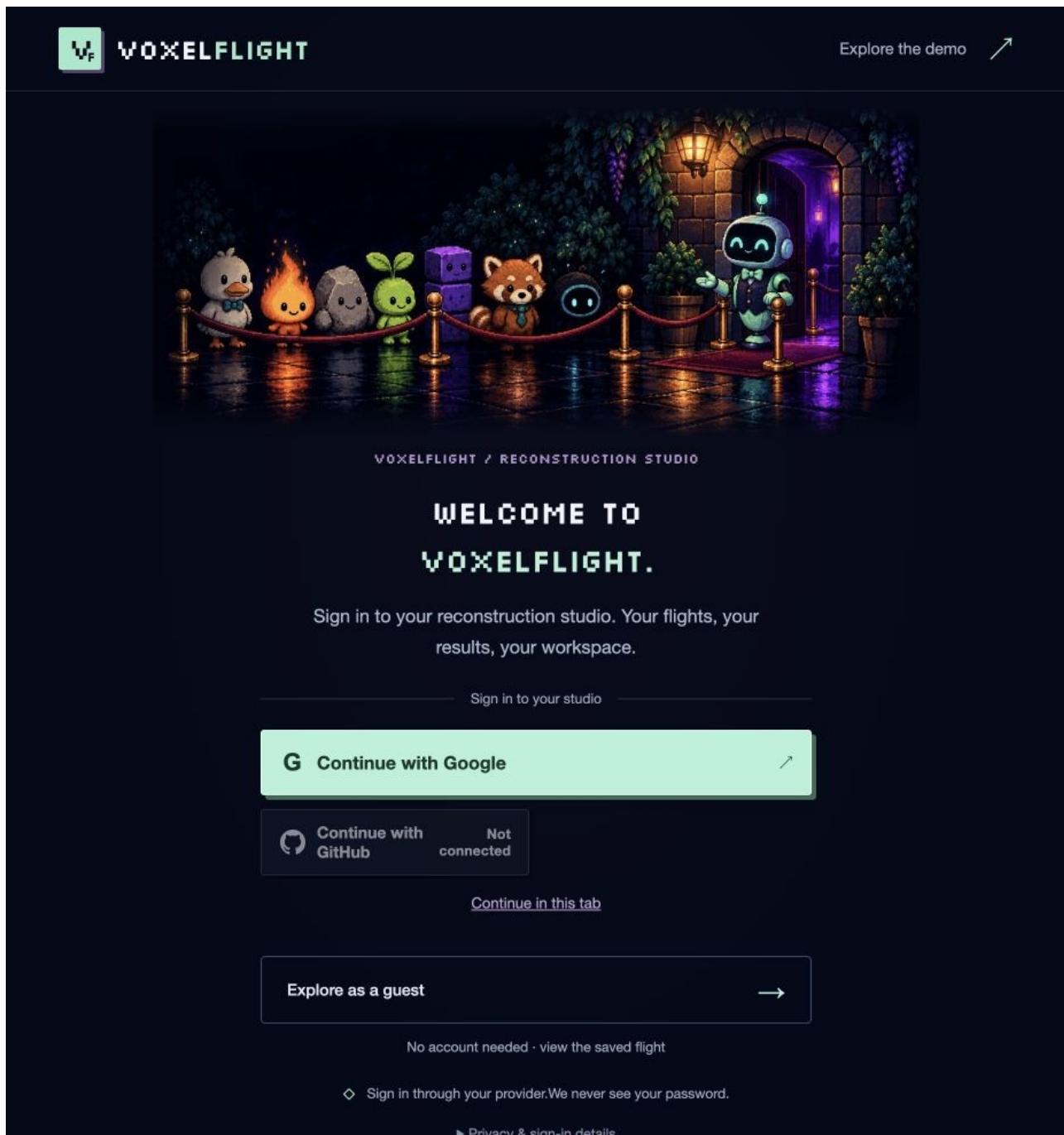
The required long-video benchmark has not been run.

Viewer and geospatial exports

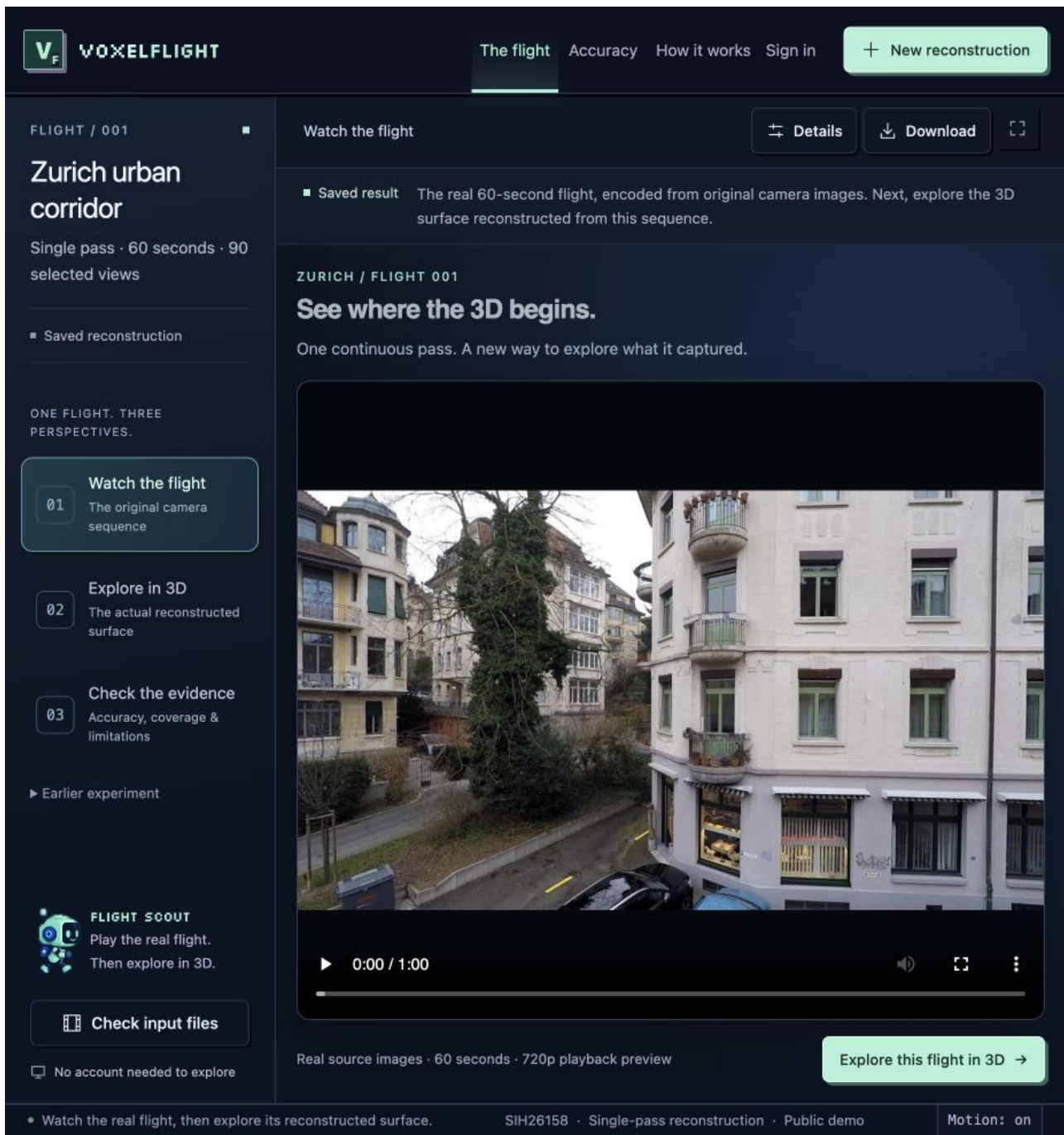
Public viewer and several formats available. FBX remains undemonstrated.

A video-first workspace for inspection

Reviewers can watch the source flight, inspect the actual geometry and download the available scene files.



Sign-in and guest access



Source flight and 3D workspace

Infrastructure inspection

Review facade and road context before a targeted follow-up survey.

Disaster damage review

Inspect visible areas remotely. Keep unseen regions explicit.

Planning and documentation

Reuse scene files with their coordinate reference information.

voxelflight-3d.web.app

Saved demo is public. New processing requires an available worker. Field benefits need pilot validation.

Research foundations and reproducible delivery

The project integrates published geometry methods with input handling, source texture and an inspection interface.

[1] COLMAP

Camera calibration, SfM and bundle adjustment

[2] MapAnything

Pretrained metric depth and camera geometry

[3] Open3D TSDF integration

Fusion of supported RGB-D observations

[4] Zurich Urban MAV

Public imagery, telemetry and reference trajectory

Geometry with observed support

Combines calibrated visual anchors with learned depth. Cross-view checks restrict fusion, and real camera images provide surface texture.

Separate evidence for each claim

Distinguishes source video, actual geometry and archived appearance renders. Reports aligned and absolute trajectory error separately.

Inspectable public implementation

Publishes source, setup instructions, evaluation, screenshots and an editable presentation.
A fixed-input benchmark remains the next test.